INCH-POUND
MIL-W-25038/3C
16 Sep 1993
SUPERSEDING
MIL-W-25038/3B
20 September 1991

MILITARY SPECIFICATION SHEET

WIRE, ELECTRICAL, HIGH TEMPERATURE, FIRE RESISTANT, AND FLIGHT CRITICAL, LIGHT WEIGHT, SMALL DIAMETER 1/

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-W-25038.

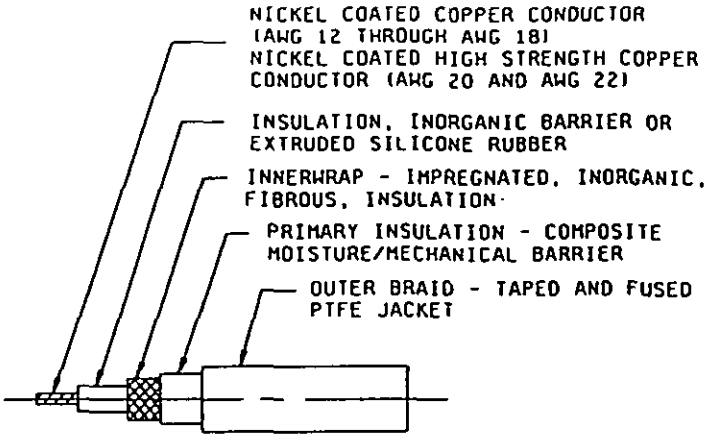


FIGURE 1. Design and construction.

REQUIREMENTS:

Design and construction: See figure 1.

Temperature rating: 260°C.

Voltage rating: 600 V rms.

1/ Size 22H wire is a heavy (thick-wall) version of size 22 wire for use in specific applications as referenced in footnotes 4/ and 5/ of table I. It is considered to be "light weight, small diameter" when compared with the normal weight size 22 of MIL-W-25038/1.

Changes to the text have been highlighted.

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Thermal shock: .060 inch maximum change in insulation measurement.

Durability of identification: 250 cycles (500 strokes) at 150 gram weight.

Color: Unless otherwise specified by the Part or Identifying Number (PIN), the color shall be white.

PIN: M25038/3- (wire size from table I)-(color code designator in accordance with MIL-STD-681).

Examples:

Size 20 AWG, white = M25038/3-20-9.
Size 20 AWG, white with orange stripe = M25038/3-20-93.

NOTE: Materials used in construction of the wire must be identified and approved by the qualifying activity. Wire construction cannot contain asbestos.

TABLE I. Finished wire construction and performance details.

Wire size 1/	Conductor strand 2/			Maximum conductor diameter (inch)	Maximum dc resistance 1,000 ft. at 20°C (ohms)	Finished wire diameter (inch) 3/		Weight lbs/1,000 ft (max) (lbs)	Performance details	
	Number of strands	AWG size	Nominal diameter (inch)						Bend test mandrel	
						Min	Max		OD (inches)	Load (lbs)
22 4/ 5/	19	34	.0063	.033	23.70	.040	.054	4.15	6	2
22H	19	34	.0063	.033	23.70	.055	.075	6.0	6	2
20	19	32	.008	.041	15.27	.048	.083	9.0	6	2
18	19	30	.010	.052	8.5	.065	.097	10.5	6	2
16	19	29	.0113	.061	6.66	.068	.103	13.5	8	3
14	19	27	.0142	.074	4.32	.097	.123	19.5	8	3
12	19	25	.0179	.093	2.78	.100	.142	28.0	8	3

1/ Wire size 20 AWG and 22 AWG shall be constructed with nickel coated high strength copper strands.

2/ The conductor strands shall have an annular nickel coating, whose weight shall not be less than 27 percent of the total weight of the nickel-coated strand when tested in accordance with ASTM B355.

3/ These wires comply with the sealing range requirements for firewall connectors as specified in MIL-C-5015, MIL-C-38999, and MIL-C-83723.

4/ The "22H" wire is defined as heavy-wall or thick-wall construction.

5/ The "22H" wire should only be used with a size 20 type contact for MIL-C-38999 connectors and comply to the sealing requirements of AIR 1329.

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CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - AS
Air Force - 85
NASA - NA

Review activities:
Army - AR, MI
Air Force - 99
DLA - ES

User activities:
Army - AV
Navy - OS

Preparing activity:
Air Force - 85

Agent:
DLA - ES

(Project 6145-2032-2)

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